Lab: Explore Signatures and Logs with Suricata

Activity Overview

In this lab, I explored Suricata, an open-source network security tool used to monitor network traffic and generate alerts based on predefined rules.

The lab involved examining a custom rule, running Suricata against a sample packet capture, and checking the generated logs. I also used `jq` to read and filter the event data stored in JSON format.

Objectives

- Examine a custom Suricata rule.
- Run Suricata against sample network traffic.
- Review alerts in `fast.log`.
- Examine and filter events in `eve.json`.

Tools and Files

- **Suricata:** Used to detect matching network traffic.
- **Bash Terminal:** Used to run commands.
- **jq:** Used to read and filter JSON data.
- **custom.rules:** Contains the detection rule.
- **sample.pcap:** Contains sample network traffic.
- **fast.log and eve.json:** Store Suricata alerts and event information.

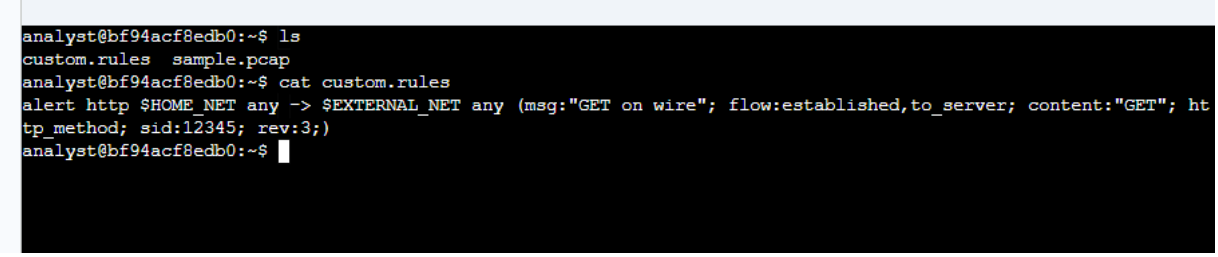
Task 1: Examine a Custom Rule in Suricata

First, I used the `cat` command to view the contents of the `custom.rules` file.

Command:

```
cat custom.rules
```

Screenshot 1: View the custom rule



This screenshot shows the custom Suricata rule displayed in the terminal.

The rule used in this lab was:

```
alert http $HOME_NET any -> $EXTERNAL_NET any (msg:"GET on wire";
flow:established,to_server; content:"GET"; http_method; sid:12345; rev:3;)
```

The rule contains three main parts: an action, a header, and rule options. The `alert` action generates an alert when the specified traffic matches the rule. The header defines the protocol and traffic direction, while the options specify the conditions for triggering the alert.

Task 2: Trigger a Custom Rule and Check the Alerts

Step 1: Check the Log Directory

I checked the Suricata log directory before running the packet capture.

Command:

```
ls -l /var/log/suricata
```

Step 2: Run Suricata

I ran Suricata using the provided packet capture and custom rules file.

Command:

```
sudo suricata -r sample.pcap -S custom.rules -k none
```

Step 3: Check the Generated Files

After running Suricata, I checked the log directory again.

Command:

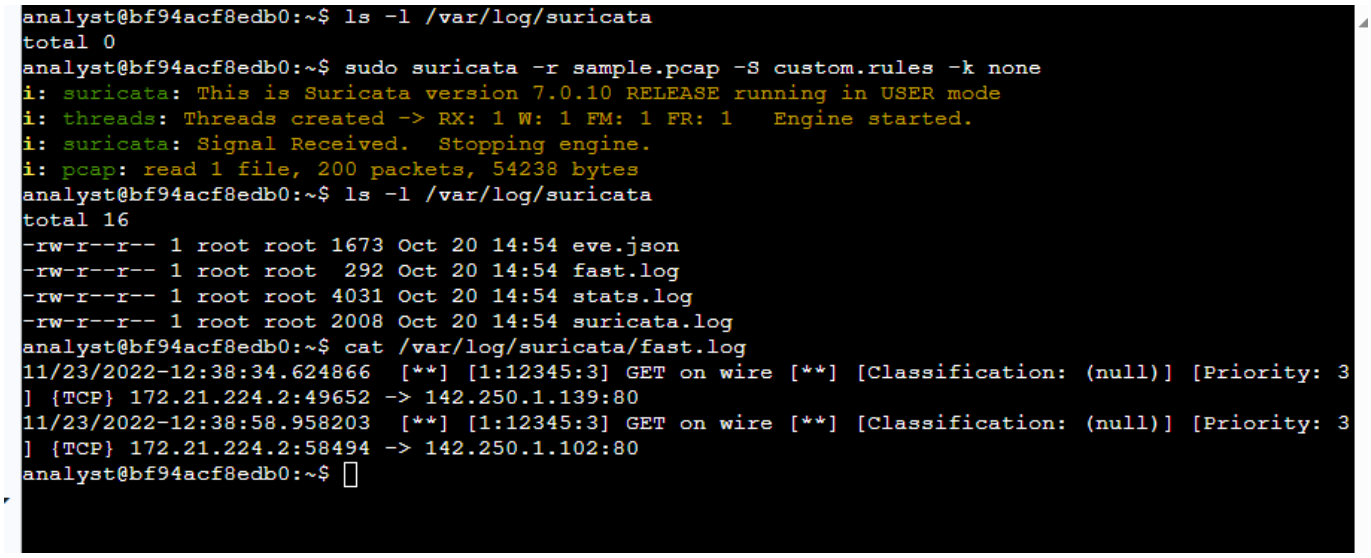
```
ls -l /var/log/suricata
```

Step 4: View the Alerts

I used the following command to display the alerts generated by Suricata.

Command:

```
cat /var/log/suricata/fast.log
```



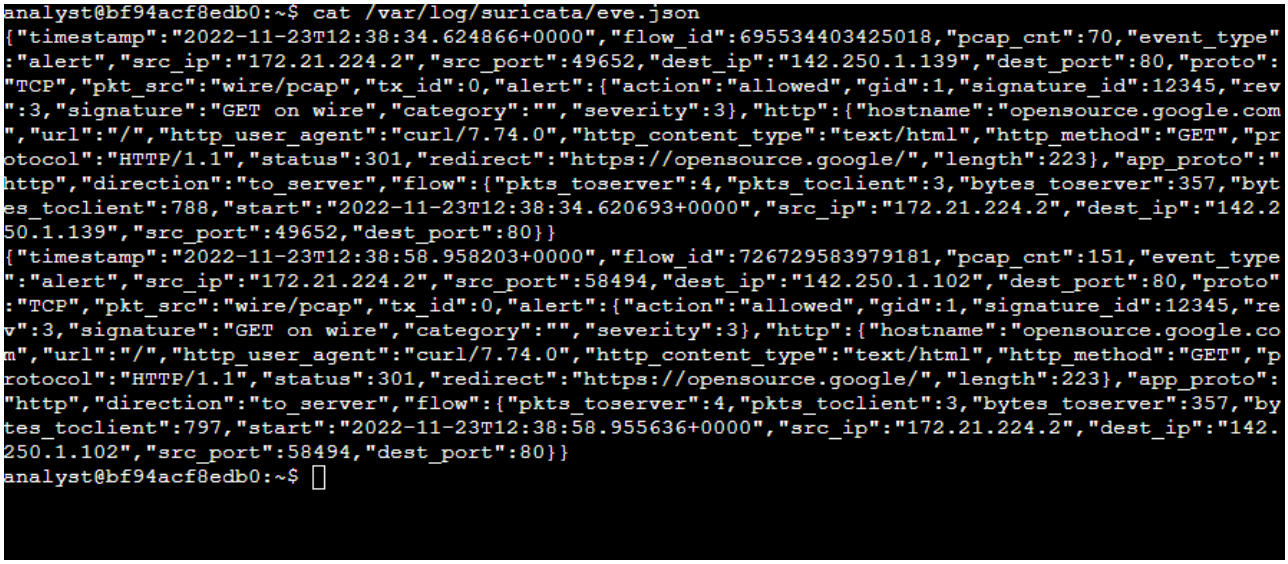
Task 3: Examine eve.json Output

Step 1: View the Event Log

I used the `cat` command to display the contents of the `eve.json` file.

Command:

```
cat /var/log/suricata/eve.json
```



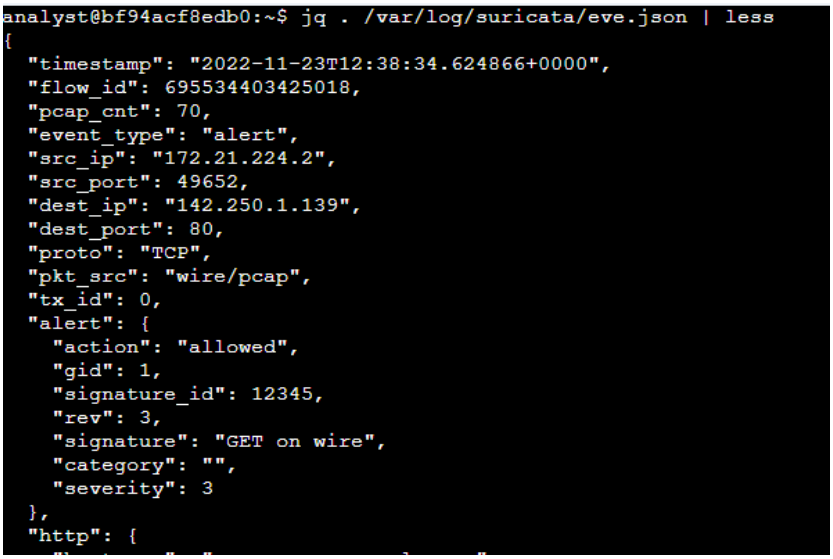
This screenshot shows the raw JSON event data generated by Suricata.

Step 2: Format the JSON Output

I used `jq` with `less` to display the event data in a more readable format.

Command:

```
jq . /var/log/suricata/eve.json | less
```



This screenshot shows the formatted event records in the terminal.

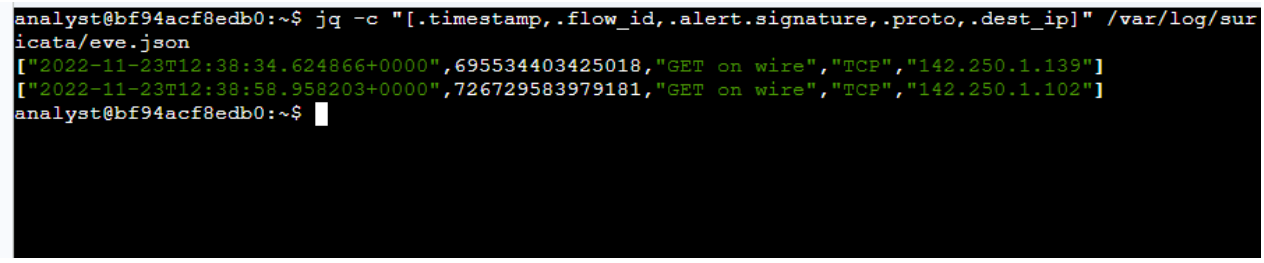
I pressed `q` to exit the `less` viewer and return to the terminal.

Step 3: Extract Specific Event Information

I used `jq` to display selected fields from the event log.

Command:

```
jq -c "[.timestamp,.flow_id,.alert.signature,.proto,.dest_ip]"
/var/log/suricata/eve.json
```



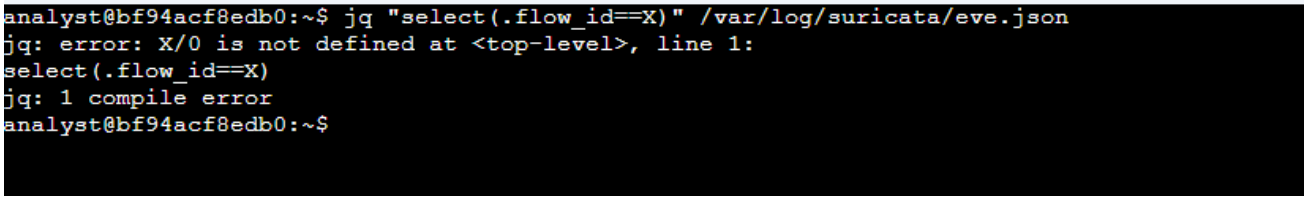
This screenshot shows the timestamp, flow ID, alert signature, protocol, and destination IP address extracted from the event log.

Step 4: Filter Events by Flow ID

I used the `flow_id` field to display events associated with a particular network flow.

Command:

```
jq "select(.flow_id==14500150016149)" /var/log/suricata/eve.json
```



This screenshot shows the event records associated with the selected flow ID.

Note: The flow ID in this command is an example. I replaced it with an actual flow ID from my own terminal output when performing the lab.

Key Learnings

- Learned how Suricata rules are structured.
- Practiced running Suricata against a packet capture.
- Viewed alerts generated in `fast.log`.
- Examined event records in `eve.json`.
- Used `jq` to extract information and filter events by flow ID.

Conclusion

This lab provided practical experience with Suricata rules and network log analysis. I learned how to process sample network traffic, view generated alerts, and examine event data using command-line tools.

These skills are useful for monitoring network activity and investigating security alerts.

Lab Evidence

The screenshots included in this document show the commands used and the results obtained during the lab activity.

Course: Google Cybersecurity Professional Certificate

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